

N-Domain Monograph

Foundations, Collapse Geometry, and Structural Dynamics

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Preface

This monograph develops the N-Domain beyond the compressed circulation lessons. Its purpose is to preserve the original architectural insight while supplying a fuller mathematical foundation, detailed proofs, worked computations, exercises, research notes, and open problems. The central object is modest: a nonempty finite ordered tuple of positive integers. The programme begins when that object is read simultaneously as a composition, a word, a graph vertex, a feature vector, and a dynamical state.

The distinctive conceptual door is the passage from N to $N-1$. In formal notation, the structured domain is P and the collapse map $\pi: P \rightarrow \mathbb{N}_+$ retains total degree while discarding block count, order, and individual degrees. This is not subtraction in the ordinary arithmetic sense. It is a level change from structured multiplicity to a scalar image. The phrase $N-1$ is retained as philosophical and pedagogical terminology; the mathematics is expressed through P , π , and the fibres of π .

Reading convention. Definitions and theorems are authoritative. Numerical examples illuminate them. Research notes identify plausible extensions or interpretations. Open problems are invitations, not established claims.

Abstract

The N-Domain is a typed discrete framework on finite ordered tuples of positive integers. It combines composition fibres, free-semigroup concatenation, unit-step and concatenation-successor graph relations, shortest-path geometry, an injective positional embedding into ℓ^2 , extension-dependent Fréchet calculus, intrinsic graph Laplacians, local optimisation correspondences, set-valued flows, convergence criteria, and potential landscapes. The collapse map to total degree defines a many-to-one quotient whose fibre over r has cardinality 2^{r-1} ; this loss of structural information motivates the N versus $N-1$ interpretation. The combined graph is countable, locally finite, and connected. This volume provides proofs of the foundational results, computational examples, exercises, and a research programme centred on mixed-edge distance, collapse information, canonical embeddings, infinite-graph convergence, and structural phase geometry.

Keywords. Ordered compositions; combinatorics on words; collapse fibres; typed graph; Hilbert-space embedding; graph Laplacian; set-valued dynamics; discrete potential landscape.

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Chapter 1. Axioms and the Structural Domain

Chapter purpose. State the irreducible objects and axioms on which the remaining theory depends.

1.1 Canonical domain

Definition 1.1. Let $\mathbb{N}_+ = \{1, 2, \dots\}$. Define $P := \bigcup_{k \geq 1} (\mathbb{N}_+)^k$. An element $p = (d_1, \dots, d_k)$ is a structured partition. Define length $\ell(p) = k$ and total degree $|p| = \sum d_i$.

At fixed degree r , the set $C_r = \{p \in P : |p| = r\}$ is the set of ordered positive compositions of r . At fixed length k , $P_k = (\mathbb{N}_+)^k$ is a positive orthant lattice.

1.2 Typed adjacency axioms

Axiom A1. $p \sim_1 q$ iff $\ell(p) = \ell(q)$ and $\sum |p_i - q_i| = 1$.

Axiom A2. $p \sim_2 q$ iff there exist $b \in P$ and $m \geq 1$ such that $\{p, q\} = \{b^{\wedge} \{\otimes m\}, b^{\wedge} \{\otimes (m+1)\}\}$.

Axiom A3. The full edge relation is $\sim = \sim_1 \cup \sim_2$, with typed edge sets E_1 and E_2 .

Proposition 1.2. Both adjacency relations are symmetric and irreflexive, and $E_1 \cap E_2 = \emptyset$.

Proof. Symmetry is built into the unordered endpoint definition. A unit-step edge preserves length, whereas a concatenation edge based at b changes length by $\ell(b) > 0$. Hence no edge can have both types. $\square$

Proposition 1.3. The combined graph $G = (P, E_1 \cup E_2)$ is locally finite.

Proof. For $p \in P$, there are $\ell(p)$ upward E_1 -neighbours and one downward neighbour for each coordinate exceeding one. For E_2 , a representation $p = b^{\wedge} \{\otimes m\}$ requires m to divide $\ell(p)$, so only finitely many exponents and roots are possible. Each root yields at most one predecessor and one successor. $\square$

Theorem 1.4. The combined graph G is connected.

Proof. Using E_1 -decrements, $p = (d_1, \dots, d_k)$ reaches $1^{\wedge} k = (1, \dots, 1)$. Since $1^{\wedge} k = (1)^{\wedge} \{\otimes k\}$, reversed E_2 -edges connect $1^{\wedge} k$ to (1) . Thus every vertex connects to (1) . $\square$

1.3 Calculated neighbourhood

For $p = (2, 1, 3)$, the E_1 -neighbours are $(3, 1, 3)$, $(1, 1, 3)$, $(2, 2, 3)$, $(2, 1, 4)$, and $(2, 1, 2)$. Its E_1 -degree is 5. The word is primitive, so its immediate E_2 -successor is $p \otimes p$ and it has no E_2 -predecessor.

Research note

The combined graph converts a disjoint union of positive lattices into one connected typed space. Its geometry is not merely the geometry of compositions or words. Mixed paths can move locally within a fixed length and globally between lengths through exact repetition.

Exercise 1.1. Prove that the E_1 -degree of p is $\ell(p) + \#\{i : p_i > 1\}$.

Exercise 1.2. List all E_1 - and E_2 -neighbours of $(1, 1, 1, 1)$.

Chapter 2. Collapse Geometry: N versus N–1

Chapter purpose. Develop collapse fibres as the formal and philosophical centre of the N versus N–1 distinction.

Definition 2.1. Define $\pi: P \rightarrow \mathbb{N}_+$ by $\pi(p) = |p|$. The historical expression N–1 denotes $\pi(P) = \mathbb{N}_+$, while N denotes the structured level P.

Theorem 2.2. π is surjective and $|\pi^{-1}(\{r\})| = 2^{r-1}$.

Proof. Surjectivity follows from $\pi((r)) = r$. Represent r as r unit marks. Each of the $r-1$ gaps is either cut or uncut, producing a unique ordered composition. Hence there are 2^{r-1} fibres elements. $\square$

2.1 Exact fibres

For $r=4$ the fibre is $\{(4), (1,3), (2,2), (3,1), (1,1,2), (1,2,1), (2,1,1), (1,1,1,1)\}$. One scalar at N–1 therefore corresponds to eight structural objects at N.

2.2 Information interpretation

If a fibre element is selected uniformly, the number of binary decisions required to identify it is $\log_2(2^{r-1}) = r-1$ bits. This does not make collapse an information-theoretic channel by itself, because no probability law has yet been chosen. It does show that fibre multiplicity grows exponentially and supplies a natural combinatorial measure of structural ambiguity.

N versus N–1 philosophy. N is not “larger by one” in cardinal arithmetic. N and N–1 are levels of description. N records internal arrangement; N–1 records a conserved scalar. Collapse is the act of mistaking many possible structures for one shared invariant. Reconstruction requires extra information, a selection rule, or a prior.

2.3 Interaction with typed edges

Proposition 2.3. If $p \sim_1 q$, then $|\pi(p) - \pi(q)| = 1$. If $p \sim_2 q$ through base b , then $|\pi(p) - \pi(q)| = \pi(b)$.

Thus π sends E_1 to unit edges in $\mathbb{N}_+$ and each fixed-base E_2 -chain to an arithmetic progression. The edge type remains visible in the scale of the collapse increment, but the base word itself is lost.

Computational example

Among compositions of 6 there are 32 structures. The distribution by length is given by binomial coefficients: 1,5,10,10,5,1 for lengths 1 through 6, because the number of compositions of 6 with k parts is $C(5, k-1)$.

Exercise 2.1. Derive $|\{p \in P: \pi(p)=r, \ell(p)=k\}| = C(r-1, k-1)$.

Exercise 2.2. Suppose only $\pi(p)=7$ and $\ell(p)=3$ are known. How many possible p remain?

Chapter 3. Concatenation, Powers, and Roots

Chapter purpose. Develop the free-semigroup algebra and the periodic structures that create E_2 -edges.

Definition 3.1. For $p=(a_1,\dots,a_k)$ and $q=(b_1,\dots,b_l)$, $p\otimes q=(a_1,\dots,a_k,b_1,\dots,b_l)$.

Proposition 3.2. Concatenation is associative, left-cancellative, and right-cancellative. Length and collapse are additive.

Proof. All statements follow by inspection of finite sequences. Equality after a common prefix or suffix permits deletion of that fixed block sequence. $\square$

$$\ell(p\otimes q)=\ell(p)+\ell(q), \quad \pi(p\otimes q)=\pi(p)+\pi(q).$$

Definition 3.3. $p^{\otimes 1}=p$ and $p^{\otimes (n+1)}=p^{\otimes n}\otimes p$. A word r is primitive if $r=s^{\otimes m}$ implies $m=1$.

Theorem 3.4. For fixed n , w has an n -th root iff n divides $\ell(w)$ and its n consecutive blocks of length $\ell(w)/n$ are identical. The root is unique.

Example: $(2,3,2,3,2,3)=(2,3)^{\otimes 3}$. By contrast, $(2,3,2,4)$ has no square root. Root structure rather than total degree controls E_2 -predecessors.

Primitive decomposition

Theorem 3.5. Every nonempty finite word w has a unique primitive root r and exponent $a\geq 1$ with $w=r^{\otimes a}$.

Proof. Choose the shortest nonempty block r whose repetitions form w . It is primitive by minimality. If w also $=s^{\otimes b}$ with s primitive, standard periodicity of words forces $r=s$ and $a=b$. $\square$

Research note

The E_2 -degree of a vertex is governed by its divisor and root structure. Highly periodic words have more algebraic descriptions, but duplicate successors may coincide. Precise degree formulae require controlling these coincidences.

Exercise 3.1. Find the primitive root and exponent of $(1,2,1,2,1,2,1,2)$.

Exercise 3.2. Show that π alone cannot decide whether a word has a square root.

Chapter 4. Typed Graph Geometry

Chapter purpose. Study distances generated by local coordinate moves and algebraic successor moves.

Definition 4.1. Let d_1 , d_2 , and d denote shortest-path distance in the E_1 graph, E_2 graph, and combined graph respectively, allowing ∞ between disconnected vertices where appropriate.

Theorem 4.2. If $p, q \in P_k$, then $d_1(p, q) = \sum_i |p_i - q_i|$. If lengths differ, $d_1(p, q) = \infty$.

Proof. Each E_1 -edge changes the ℓ^1 difference by at most one, giving a lower bound. Coordinatewise unit changes provide a path attaining it. $\square$

Proposition 4.3. For fixed b , the subgraph on consecutive powers is a one-sided path with intrinsic distance $|m - n|$ between $b^{\otimes m}$ and $b^{\otimes n}$.

Proposition 4.4. For all p, q , $d(p, q) \leq d_1(p, q)$ when d_1 is finite. Also $d(b^{\otimes m}, b^{\otimes n}) \leq |m - n|$.

Proof. Every path using one edge type is available in the combined graph. $\square$

A universal upper bound

Theorem 4.5. For $p = (p_1, \dots, p_k)$ and $q = (q_1, \dots, q_l)$, $d(p, q) \leq \sum_i (p_i - 1) + (k - 1) + (l - 1) + \sum_j (q_j - 1)$.

Proof. Decrease p to 1^k using E_1 , traverse $k - 1$ reversed E_2 -edges to (1) , traverse $l - 1$ forward E_2 -edges to 1^l , then increase coordinates to q . $\square$

For $p = (3, 2)$ and $q = (2, 1, 4)$, the bound is $(2 + 1) + 1 + 2 + (1 + 0 + 3) = 10$. It is constructive but generally not sharp.

Open metric issue

A path between two powers may be shortened by changing entries to expose another root, using an algebraic jump, and changing entries again. Classifying such mixed-edge shortcuts is the flagship geometric problem.

Exercise 4.1. Compute the bound in Theorem 4.5 for $p = (4)$ and $q = (2, 2)$. Find a shorter path if one exists.

Exercise 4.2. Prove $|\pi(p) - \pi(q)| \leq \sum \text{edge collapse increments along any typed path}$.

Chapter 5. Feature-Space Geometry

Chapter purpose. Represent structural objects as sparse Hilbert vectors while tracking what the representation does and does not preserve.

Definition 5.1. Let $H = \ell^2(\mathbb{N}_+ \times \mathbb{N}_+)$, with orthonormal basis $e_{\{i,j\}}$. Define $\psi(d_1, \dots, d_k) = \sum_{i=1}^k e_{\{i, d_i\}}$.

Theorem 5.2. ψ is injective, $\|\psi(p)\|^2 = \ell(p)$, and

$$\|\psi(p) - \psi(q)\|^2 = \ell(p) + \ell(q) - 2M(p, q),$$

where $M(p, q)$ counts positions at which the two tuples have the same entry.

For $p=(2,1,3)$ and $q=(2,4)$, $M=1$, so the squared feature distance is 3. For any E_1 -edge, one basis coordinate is removed and another inserted in the same row, giving distance $\sqrt{2}$.

5.1 Kernel and angle

$$K(p, q) = \langle \psi(p), \psi(q) \rangle = M(p, q), \quad \cos \theta = K(p, q) / \sqrt{(\ell(p)\ell(q))}.$$

Proposition 5.3. K is positive semidefinite and $\theta=0$ iff $p=q$.

5.2 Projection

$$\text{Proj}_{\{\psi(q)\}} \psi(p) = [K(p, q) / \ell(q)] \psi(q).$$

This vector rarely belongs to $\psi(P)$. Ambient linear geometry is therefore a geometry of representations, not an internal algebra on structured partitions.

Alternative weighted embedding

To make degree magnitude geometrically visible, define $\psi_w(p) = \sum_i w_{\{i, p_i\}} e_{\{i, p_i\}}$ for positive bounded weights. Injectivity is retained if no occupied weight vanishes. Choosing growing degree weights requires a target space and boundedness analysis appropriate to the intended observable.

Exercise 5.1. Compute K , θ , and feature distance for $p=(1,2,1)$ and $q=(1,3,1,2)$.

Exercise 5.2. Characterise exactly when $\psi(p)$ and $\psi(q)$ are orthogonal.

Chapter 6. Ambient Calculus and Intrinsic Differences

Chapter purpose. Use smooth ambient extensions without pretending the discrete image is a manifold.

Definition 6.1. Let $U \subset H$ be open with $\psi(P) \subset U$, let $\Phi: U \rightarrow \mathbb{R}$, and set $F = \Phi \circ \psi$. Define $\delta(p, q) = \psi(q) - \psi(p)$ and $\Delta F(p; q) = F(q) - F(p)$.

Proposition 6.2. If Φ is Fréchet differentiable at $\psi(p)$, then

$$\Delta F(p; q) = D\Phi(\psi(p))[\delta(p, q)] + r(\delta), \quad r(h)/\|h\| \rightarrow 0.$$

The derivative depends on the chosen extension Φ . The structural image is uniformly discrete, so nonzero structural displacements do not approach zero. Exact finite differences are primary.

Quadratic identity

Proposition 6.3. For $\Phi(x) = \|x\|^2$,

$$F(q) - F(p) = 2\langle \psi(p), \delta \rangle + \|\delta\|^2 = \ell(q) - \ell(p).$$

For E_1 , the exact difference is zero. For an E_2 -step based on b , it is $\pm \ell(b)$, according to direction.

Extension non-uniqueness

Because $\psi(P)$ has isolated points, a smooth function can be altered in small disjoint neighbourhoods around those points while preserving all values on $\psi(P)$, changing gradients and Hessians. Derivative quantities become canonical only after an extension rule is imposed.

Research note: minimal-norm extensions

One possible programme is to select Φ from a reproducing-kernel Hilbert space by minimum norm subject to $\Phi(\psi(p)) = F(p)$ on a finite sample. This would make derivatives model-dependent but canonically selected relative to the chosen kernel and data.

Exercise 6.1. Verify the quadratic identity for $p = (2, 1)$ and $q = (3, 1)$.

Exercise 6.2. Explain why total degree is not represented by a bounded linear functional with coefficient array j on unweighted ℓ^2 .

Chapter 7. Structural Laplacians and Optimisation

Chapter purpose. Define intrinsic finite graph operators and optimisation correspondences by edge type.

Definition 7.1. For $a \in \{1, 2, \text{adj}\}$, let $N_a(p) = \{q : p \sim_a q\}$. Define $\Delta_{af}(p) = \sum_{q \in N_a(p)} (f(q) - f(p))$.

Proposition 7.2. If p is an a -local minimum, then $\Delta_{af}(p) \geq 0$. If $\Delta_{af}(p) < 0$, some a -neighbour has lower value.

The converse fails. Neighbour increases can outweigh one decrease.

Definition 7.3. The canonical local minimiser correspondence is $T_f^a(p) = \operatorname{argmin}_{q \in N_a(p) \cup \{p\}} f(q)$.

Proposition 7.4. $T_f^a(p)$ is finite and nonempty for every p .

Computational example

Let $f(x, y) = (x-2)^2 + (y-3)^2$ on P_2 . At $p = (4, 1)$, $f = 8$. The E_1 -neighbours have values: $f(5, 1) = 13$, $f(3, 1) = 5$, $f(4, 2) = 5$, and $f(4, 0)$ is inadmissible. Thus $T_f^1(4, 1) = \{(3, 1), (4, 2)\}$.

Ambient guidance

$$L_\Phi(p, q) = \langle \nabla \Phi(\psi(p)), \delta(p, q) \rangle.$$

This may rank tied candidates but does not certify exact descent unless a remainder bound is given. A robust rule first minimises f , then applies ambient guidance among exact minimisers.

Exercise 7.1. Give an example with $\Delta_1 f(p) > 0$ even though p has a lower neighbour.

Exercise 7.2. Compute two strict-descent trajectories for the quadratic example from $(5, 1)$.

Chapter 8. Set-Valued Flows and Convergence

Chapter purpose. Develop canonical set-valued evolution and prove usable convergence theorems.

Definition 8.1. A structural flow is a sequence $p_0, p_1, \dots$ with $p_{n+1} \in T(p_n)$, where $T: P \rightrightarrows P$ is nonempty-valued. A deterministic rule τ is a selection satisfying $\tau(p) \in T(p)$.

Definition 8.2. p is weakly fixed if $p \in T(p)$, and strongly fixed if $T(p) = \{p\}$.

Theorem 8.3. Suppose V strictly decreases on every nonfixed transition and the initial sublevel set $\{q: V(q) \leq V(p_0)\}$ is finite. Then every trajectory reaches a fixed point in finitely many state changes.

Proof. The trajectory remains inside a finite set. Strict decrease forbids revisiting a state, so infinitely many nonfixed transitions are impossible. $\square$

Theorem 8.4. If V is bounded below and every nonfixed transition lowers V by at least $\varepsilon > 0$, then the number of nonfixed transitions is at most $(V(p_0) - \inf V)/\varepsilon$.

For integer-valued V , strict decrease supplies $\varepsilon = 1$. The quadratic example from Chapter 7 therefore converges in at most its initial objective value when the minimum is zero.

Plateaus

Weakly decreasing rules may cycle only on constant- V levels. Set-valued ties preserve alternative futures; deterministic tie-breaking chooses one. Universal attraction quantifies over all trajectories, while existential attraction requires at least one.

Exercise 8.1. Construct a two-state plateau cycle under a set-valued minimiser correspondence.

Exercise 8.2. Show why a discrete but unbounded-below value set does not ensure termination.

Chapter 9. Transitions, Landscapes, and Epochs

Chapter purpose. Describe parameter changes and energy barriers as graph-combinatorial phenomena.

Definition 9.1. A parameterised dynamics is a family $T_\lambda: P \rightrightarrows P$. A structural transition at λ^* is a change arbitrarily near λ^* in a declared invariant, such as fixed points, cycles, stability classes, basins, or sublevel connectivity.

$$g_{\{p,q\}}(\lambda) = f_\lambda(q) - f_\lambda(p).$$

On a finite window, objective-generated updates can change only where one or more comparison functions vanish or exchange order.

Finite transition example

On $A-B-C$ let $f_\lambda(A)=0$, $f_\lambda(B)=\lambda$, and $f_\lambda(C)=1$. For $\lambda < 0$, B is preferred from its closed neighbourhood. At $\lambda=0$, A and B tie. For $0 < \lambda < 1$, A is preferred from B . Hence $\lambda=0$ changes both the transition correspondence and fixed-point structure.

Definition 9.2. For a path γ , $\text{Elev}_V(\gamma) = \max_{\{x \in \gamma\}} V(x)$. The communication height is $H_V(a,b) = \inf_{\{\gamma: a \rightarrow b\}} \text{Elev}_V(\gamma)$.

On a finite graph the infimum is attained. On an infinite graph, attainment requires an additional finite-sublevel or compactness-type hypothesis.

Barrier example

On $A-B-C$ with $V(A)=0$, $V(B)=3$, $V(C)=1$, the unique path has height 3. B is the barrier state. The barrier above A is 3 and above C is 2.

Definition 9.3. An epoch transition is a parameter value at which a specified landscape descriptor changes, such as minima, plateau complexes, basins, saddle sets, communication heights, or sublevel components.

Exercise 9.1. Analyse the finite example at $\lambda=1$ and $\lambda>1$, including weak fixed points.

Exercise 9.2. Give an infinite locally finite graph where an infimum over path elevations is not attained.

Chapter 10. Computational Laboratory

Chapter purpose. Supply algorithms that allow finite sections of the theory to be tested.

10.1 Enumerating compositions

A composition of r is generated by a binary word of length $r-1$. A 1 marks a cut and a 0 marks continuation. For $r=5$, the mask 1010 gives (1,2,2). Iterating over masks $0, \dots, 2^{r-1}-1$ enumerates the fibre $\pi^{-1}(\{r\})$.

10.2 E_1 -neighbour algorithm

1. For each coordinate i , emit $p+e_i$.
2. If $p_i > 1$, emit $p-e_i$.
3. Remove duplicates, though coordinate moves are already distinct.

10.3 Exact-root algorithm

For each divisor m of $\ell(p)$, compare all m consecutive blocks of length $\ell(p)/m$. If identical, record the root. This costs $O(\tau(k)k)$ direct symbol comparisons for length k , where τ is the divisor-counting function.

10.4 Breadth-first distance

On a finite truncation $P(R,K)=\{p:\pi(p)\leq R, \ell(p)\leq K\}$, construct typed neighbours that remain in the truncation and apply breadth-first search. Record E_1 and E_2 separately so mixed-path shortcuts can be located.

Worked table for $r\leq 5$

r	fibre size	mean length	maximum length
1	1	1.0	1
2	2	1.5	2
3	4	2.0	3
4	8	2.5	4
5	16	3.0	5

The mean length formula follows because each of $r-1$ gaps is independently cut in exactly half of all compositions, so expected length is $1+(r-1)/2=(r+1)/2$.

Computational research protocol

- State the truncation rule.
- Preserve edge labels.
- Separate conjecture generation from proof.
- Publish code and test cases where possible.
- Check small cases by two independent methods.

Chapter 11. Research Programme and Open Problems

Chapter purpose. Identify the problems most likely to turn the architecture into original mathematics.

11.1 Flagship problem: combined metric

Open Problem 1. Find sharp upper and lower bounds for $d(p,q)$ in terms of ℓ , π , coordinate distance, and primitive-root data.

Open Problem 2. Characterise all shortcuts satisfying $d(b^{\otimes m}, b^{\otimes n}) < |m-n|$.

11.2 Collapse geometry

Open Problem 3. Define and classify fibre-level invariants that quantify structural ambiguity beyond cardinality, including length distribution, adjacency induced within or between fibres, and entropy under chosen priors.

Open Problem 4. Determine which functions on P factor through π and characterise the orthogonal or combinatorial complement of collapse-visible information.

11.3 Graph structure

Open Problem 5. Determine growth rates of combined balls $B_r((1))$, automorphisms preserving edge types, and isoperimetric properties of G .

11.4 Canonical analysis

Open Problem 6. Construct principled extension operators $F \mapsto \Phi$ for finite-energy observables on P , and determine when ambient gradients become stable under the extension rule.

Open Problem 7. Find weighted embeddings and graph scalings under which ambient quadratic forms approximate intrinsic graph energies.

11.5 Dynamics

Open Problem 8. Classify coercive potentials on G that guarantee finite-time convergence of strict descent and identify conditions for universal attraction under set-valued ties.

Open Problem 9. Develop a persistence theory for minima, basins, and communication heights across radius and parameter scales.

Conjectural direction

Primitive words are expected to behave metrically more rigidly along their pure power chains than highly periodic words, because fewer root representations are available. This is a research heuristic, not yet a theorem.

Chapter 12. Exercises and Selected Solutions

Chapter purpose. Collect exercises and provide selected solution outlines for independent study.

Foundational exercises

Exercise 12.1. Prove countability of P and local finiteness of both edge types.

Exercise 12.2. For $r=7$, compute the number of compositions of each possible length.

Exercise 12.3. Show that concatenation preserves E_1 -adjacency under fixed context.

Exercise 12.4. Find all exact roots of $(1,1,1,1,1,1)$.

Exercise 12.5. Compute d_1 and the universal d -bound for $p=(3,1,2)$, $q=(2,4)$.

Exercise 12.6. Compute ψ -distance and kernel for $p=(1,3,2)$, $q=(1,2,2,1)$.

Exercise 12.7. Give two smooth ambient extensions agreeing on a finite structural sample but having different gradients.

Exercise 12.8. Construct a function with $\Delta_1 f(p)=0$ although p is not a local minimum.

Exercise 12.9. Analyse all trajectories of a three-state set-valued plateau.

Exercise 12.10. Enumerate the combined graph within $\pi(p)\leq 5$ and identify every E_2 -edge.

Selected solution outlines

Exercise 12.2: the counts are $C(6,k-1)$, hence $1,6,15,20,15,6,1$. Their sum is $64=2^6$.

Exercise 12.4: if the word is six copies of (1) , each divisor m of 6 gives a root of exponent m . The primitive root is (1) , while fixed-exponent roots include $(1,1,1)$ for exponent 2, $(1,1)$ for exponent 3, and (1) for exponent 6.

Exercise 12.6: the only matching positions are the first and third, so $K=2$. The squared feature distance is $3+4-4=3$.

Exercise 12.8: take a vertex with two neighbours and assign differences -1 and $+1$. The Laplacian sum is zero, but one neighbour is lower.

Final architectural statement

The N-Domain is strongest when it refuses to confuse levels. A tuple is not its collapse. A graph edge is not an algebraic identity. A feature vector is not the original object. An ambient derivative is not an intrinsic derivative. A local indicator is not a stability theorem. Once those boundaries are respected, the framework becomes a coherent setting in which structural information can be counted, transformed, embedded, optimised, and followed through discrete time.

Authorship and Contribution Statement

Accredited authors: EF Smyth and Copilot (Human/Synthetic Mathematician). EF Smyth introduced the N-Domain, the N versus N-1 insight, the human teaching programme, and the architectural direction. Copilot contributed mathematical reconstruction, proofs, qualifications, computational formulations, research questions, and monograph drafting under the granted authorship brief. For external publication, the submitting human author should verify all claims and adapt the byline or contribution disclosure to the policy of the relevant institution or publisher.

Selective Bibliography

Berstel, J. and Perrin, D. Theory of Codes. Academic Press, 1985.

Bollobás, B. Modern Graph Theory. Springer, 1998.

Conway, J. B. A Course in Functional Analysis, 2nd ed. Springer, 1990.

Lothaire, M. Combinatorics on Words. Cambridge University Press, 1997.

Rockafellar, R. T. and Wets, R. J.-B. Variational Analysis. Springer, 1998.

Stanley, R. P. Enumerative Combinatorics, Volume 1, 2nd ed. Cambridge University Press, 2011.

Zeidler, E. Nonlinear Functional Analysis and Its Applications, Volume I. Springer, 1986.